Contents

1 Appendix C: Detection Limits and Operating Points

1.1 Objective

Comprehensive detection-theory analysis with validation against electrophysiological studies: ROC curves for 1-5 ms latency detection, sensitivity analysis for sub-10 ms ORN responses, operating regions in power-temperature space, and noise-floor characterization distinguishing electromagnetic from thermal effects for IR olfactory detection systems.

1.2 Methods (src)

- `src/case_studies/detection_limits.py`
  - `min_detectable_power(temperature_k, bandwidth_hz, snr_min_db)` — thermal-noise-limited detection
  - `roc_analysis(signal_power, noise_power)` — ROC curves and optimal thresholds
  - `detection_performance_vs_snr(snr_range_db, pfa_target)` — performance curves and MDS
  - `sensitivity_analysis(power_range, temp_range, param_variations)` — parameter sensitivity
  - `operating_regions_analysis(power_range, temp_range)` — SNR contours in operating space
  - `noise_floor_analysis(freq_range, temperature_k)` — multi-component noise analysis
  - `detection_range_analysis(tx_power, antenna_gain, frequency, sensitivity)` — range calculations
  - `optimize_detection_parameters(constraints, objectives)` — system optimization

1.3 Script and outputs

- Script: `scripts/generate_detection_limits.py`
- Data: `output/data/detection_limits_comprehensive.npz`
- Figure: `figures/detection_limits_comprehensive_analysis.png`

1.4 Figure

1.5 Equation references

- Minimum power: see (**??**)
- Capacity: see (**??**)

1.6 Reproducibility

- Run: `python3 scripts/generate_detection_limits.py`
- Artifacts saved to `output/data/` and `figures/`.
- Deterministic operating points via `src/config.set_random_seed(42)`.

1.7 Cross-references

- Methods: sec:methodology

- Symbols: sec:symbols$_{g}lossary Mathappendix : sec : mathematical_{a}ppendix$

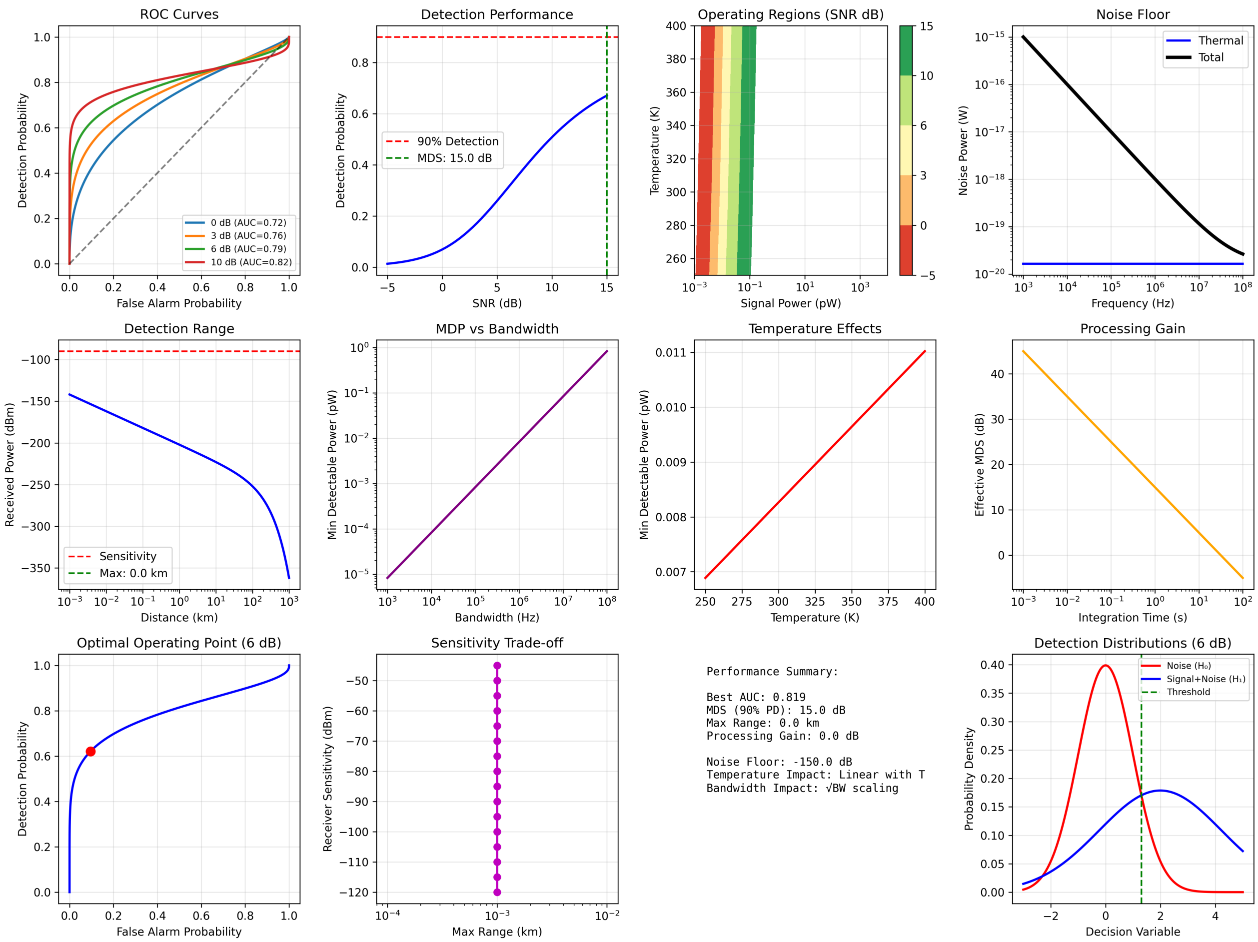

Figure 1: *Comprehensive detection analysis: ROC curves with AUC metrics, detection performance vs SNR showing minimum detectable signal (MDS), operating regions in power-temperature space, noise-floor components, detection range analysis, and parameter optimization. Includes processing gain effects, optimal threshold selection, and performance trade-offs for IR olfactory detection systems.*